Gender difference in problem solving based on electroencephalogram (EEG) signals

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Abstract*—Problem solving is regarded as one of the important skills required at the workplace and especially demanded by the industry. However, an individual's problem solving skills might differ from one another due to several factors, such as their genetic background and experiences. Hence, this preliminary study used electroencephalography (EEG) signal to investigate the differences in problem solving skills among the Malaysian undergraduates based on their gender differences. A total of 29 undergraduate students from the Faculty of Electrical Engineering, Universiti Teknologi Malaysia (UTM) participated in several experiments to complete this research. Specifically, 16 female and 13 male subjects were given two main tasks: mental arithmetic task and Tower of Hanoi (TOH) task. The EEG data were analyzed using partial directed coherence (PDC) and power spectrum estimation (PSE). Based on the results, female subjects achieved only 1% higher performance in mental arithmetic task while male subjects achieved about 13% higher performance in TOH task. The differences in terms of the functional connectivity between brain regions (in PDC) as well as the power distribution of 6 EEG waveforms, i.e. delta, theta, alpha, beta, gamma and high gamma bands are also highlighted and represented graphically in this paper.

Keywords*—problem solving, EEG, gender differences, Partial Directed Coherence (PDC), power spectrum estimation (PSE).

I. Introduction

Problem solving is a cognitive process of the brain that searches or infers a solution for a given problem in the form of a set of paths to reach a set of expected goals [1]. Individuals face problems in their everyday life. For instance, an employee has to solve problems related to their job and a student has to answer questions given by the lecturers during the class or examinations. Many surveys have been conducted on employers to find out the criteria demanded by the industry when hiring a worker. For example, the Association of American Colleges and Universities (AAC&U) has conducted a national survey on business and non-profit leaders in Washington, DC. They concluded that nearly all employers surveyed (93%) agreed that the ability to solve complex problems is more important than a candidate's undergraduate major" [2].

It is a fact that men and women differ genetically or physically. For example, a scientific finding states that the average male human brain weighs about 10% more than the female human brain, while some areas in the female human brain contain more neurons than men's [3]. Besides, some researcher claims that women have a denser corpus callosum than men which causes women being typically more emotional compare to men [4]. When relating to the brain cognitive processes, however, it is believed that the physical factor of the brain of these different sexes is not much an effector.

Consequently, in terms of the gender thinking ability, male and female might have the same performance but mentally going through different information processing path [3,4]. Moreover, through numerous experiments, it is found out that female could perform better in some tasks compared to male. For instance, women are more multitasking, good in autobiographical memory and verbal fluency [4,5]. Men on the other hand can outperform women in mathematical reasoning and visuospatial task [4,5]. Furthermore, the differences of male and female can also be accessed through their creativity [6].

EEG is one of technique used to measure the electrical activity of the brain at different sites of the head. This is typically using electrodes placed on the scalp, i.e. by using cap electrode. One of its main benefits over other brain signal recording techniques are its high temporal resolution and the fact that it can be recorded noninvasively [7]. EEG is a very accessible and useful tool for the analysis of high level brain processes since it is relatively low cost and widely used both in clinical settings and research laboratories [7]. Therefore, this study intends to overlook the gender differences focusing on problem solving task as it is one of demanded aspect in nowadays life and should be paid more focus in order to improve one performance in this aspect.

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II. Methodology

A. Population and Sample

A number of experiments have been conducted with a total of 29 participants. In general, subjects were 1st and 4th year undergraduate students with 16 female and 13 male subjects from Faculty of Electrical Engineering, Universiti Teknologi Malaysia.

B. Problem Solving Task

A problem can be classified into three types: puzzle problem, well-structured problems and ill-structured problems [8,9]. Tower of Hanoi (TOH) is in the category of puzzle problem which used to be the traditional task in cognitive science [10]. It is also a well-structured problem with close ended solution, and hence require the use of logical thinking [9]. TOH is a puzzle of basically three rods with rings. In TOH problem, one is required to move all the rings from the left-most rod to the right-most rod. The rules state that the ring could only be taken one a time and a bigger ring cannot be put on a smaller ring. Although TOH is a classic puzzle, some people nowadays might still be considered a novice in this puzzle if they never solve this kind of puzzle before.

C. Measurement

The experiment measurement can be illustrated as in Figure 1 below.

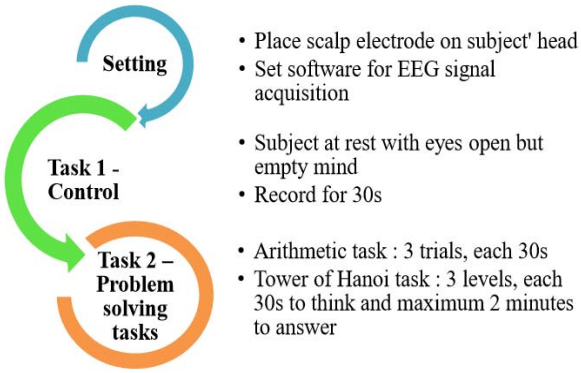

Fig. 1. Experimental measurement procedure.

First of all, subjects were given a brief explanation of the experimental procedure. The gender and year of study of the subjects were noted and recorded. Next, the scalp EEG electrode was put on the subject head and EEG signals were recorded using the NeuroFax EEG 9000 signal acquisition software. For the control task, subjects were required to be at rest and relax, and if possible with emptied mind. The subjects were asked to open their eyes during this task since the following tasks will also be recorded with open eyes. The control task was recorded for 30s for one to three times depending on a subject' mental condition (like nervousness) before they proceed with the problem solving task

Next, the problem solving task consists of two sub tasks: the arithmetic task and Tower of Hanoi task. In the arithmetic task, subjects were asked to mentally subtract 7 from 1000 continuously within 30s time. The final answer for each trial was noted and then the process was repeated for three trials after resting for about 30s per trial. EEG signals were recorded when the subjects were thinking.

Finally, the TOH task was divided into three difficulty levels: easy, medium and hard. The levels were chosen based on the feedback from subjects of preliminary experiment. EEG signals were recorded when subjects were thinking (30s) and answering (maximum two minutes) the puzzle. The rest time interval for each level was also 30s. Overall, 10 EEG data (.csv) were recorded for each subject and the recorded EEG data were named using short form of the tasks name as shown in Figure 2.

#c	= control task
#a1	= arithmetic task trial 1
#a2	= arithmetic task trial 2
#a3	= arithmetic task trial 3
#et	= TOH task easy level (think)
#ea	= TOH task easy level (answer)
#mt	= TOH task medium level (think)
#ma	= TOH task medium level (answer)
#ht	= TOH task hard level (think)
#ha	= TOH task hard level (answer)

Fig. 2. Code for recorded EEG data.

III. Findings

In this study, analyses of data were done based on two outputs from the experiment, i.e. the tabulated answer and performance of subjects and the recorded EEG data (.csv) during the tasks. The tabulated answer and performance of subjects refer to the subjects' ability to answer the given arithmetic questions with accurate final value for each trial, and also their ability to complete the TOH task within maximum 2 minutes' time provided for each level. Based on this, comparison for gender factors can be made by calculating the statistic of high performance versus low performance subjects as shown in Table 1 and further represented in graphical form as in Figure 3.

TABLE I. SUBJECTS' PERFORMANCE STATISTICS. (A) FEMALE (B) MALE

Tasks	4FRH	4FLH	1FRH	1FLH	ΣF	%	Overall	%Overall
a1	3	2	2	2	9	56.25	30	62.5
a2	2	2	2	4	10	62.5		
a3	3	4	2	2	11	68.75		
ea	4	4	4	4	16	100	28	58.33333
ma	4	3	2	2	11	68.75		
ha	0	0	0	1	1	6.25		

(a)

Tasks	4MRH	4MLH	1MRH	1MLH	ΣM	%	Overall	%Overall
a1	4	1	2	1	8	61.53846	24	61.53846
a2	4	0	2	1	7	53.84615		
a3	3	2	3	1	9	69.23077		
ea	4	2	4	3	13	100	28	71.79487
ma	3	1	4	3	11	84.61538		
ha	2	0	1	1	4	30.76923		

(b)

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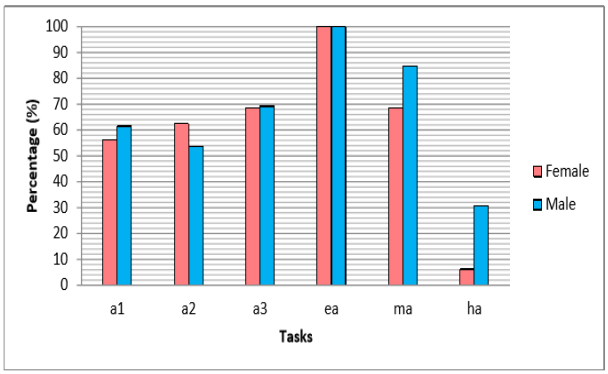
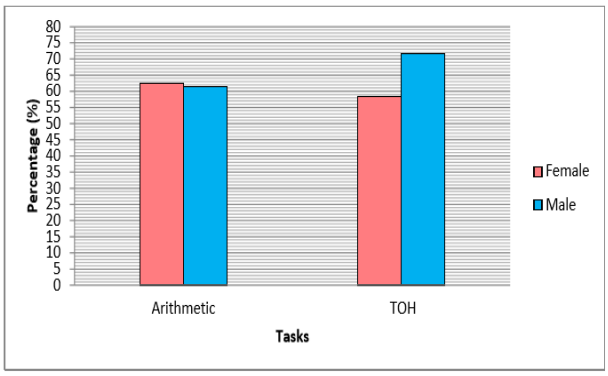


(a)



(b)

Fig. 3. Graph of gender performances (%) against tasks. (a) All trial. (b) Overall tasks.

In contrast, the recorded EEG data were analyzed using two methods, namely power spectrum estimation (PSE) and partial directed coherence (PDC) methods. The power spectrum estimation method analyse the change in power distribution of the EEG waveforms whereas the PDC analysis provides a more visual comparison based on the information pathway between the channels or brain regions of subjects when doing the given tasks. In this project, both the PSE and PDC outputs were obtained on C-program running on a Linux platform. Finally, the outputs were further analyzed using Microsoft Excel (for t-test) and Paint (for drawing).

IV. Discussions

A. PDC Analysis

The PDC output was plotted in the form a 19x19 matrix layouts of all the 19 channels used during EEG data recording. By referring to the matrix layouts as well as the three PDC values given out, the results were transformed into drawing of the information pathways for better visualization as in Figure 4.

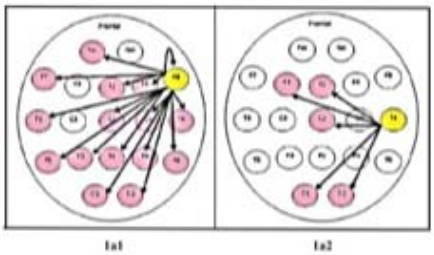


Fig. 4. Functional connectivity diagram for respondent 1 of task a1 and a2

Gender differences were determined by comparing the female and male subjects' task performance with the source of information pathway they used for the task with high performance. For each high task performance result, the source of information flow of EEG signals was identified, tabulated and can be summarized tables are shown in Table II.

Basically, the information pathway of problem solving should be originated mainly from the frontal lobe [18]. The tabulated PDC results have also shown that most of the high performance information flow sources for both the male and female are from the frontal (F) regions. Moreover, female subjects outperformed the male subjects in arithmetic task while in opposite, male subjects outperformed female subjects in TOH task. The summarized PDC source of information flow as shown in Table II (b) especially proved that the female subjects used more frontal regions in arithmetic task, while the male subjects used more frontal regions during TOH task. This is as if the utilization of the frontal regions as the source of EEG information flow during a task is proportional to the performance result of the subjects.

TABLE II GENDER DIFFERENCES IN PDC SOURCES OF INFORMATION PATHWAY FOR HIGH TASK PERFORMANCE. (A) ALL TRIAL. (B) OVERALL TASKS

	a1	a2	a3	et	ea	mt	ma	ht	ha
F	Fp1, Fp2, F8	F7, F8	Fp2, F3, F7, F8, Fz	Fp1, Fp2, F7, F8	Fp1, Fp2, F7, F8	Fp1, F7, F8	Fp1, F7, F8	F7	F7
	T3, T4, T5,T6	T3, T4, T5	T3, T4, T5	T3, T4, T5	T3, T4, T5	T3, T4, T5	T3, T4, T5, T6		T3
		O1	O2	O1, O2	O2	O2	O1		
			Pz	Pz	Pz	Cz			
M	a1	a2	a3	et	ea	mt	ma	ht	ha
	Fp1, Fp2, F7, Fz	Fp1, Fp2,F7	Fp1, F4, F7, Fz	Fp1, Fp2, F7, F8	Fp1, F7	F4, F8	Fp1, F7	F8	
	T3	T3, T5, T6	T3, T4	T3, T4	T3, T4, T6	T3, T4, T6	T3, T4, T6	T3, T4	T3, T4
	O2	O1	O2	O2	O2				
		Pz				Pz		Pz	Pz

(a)

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Arithmetic Task		TOH Task	
Female	Male	Female	Male
Fp1, Fp2, F3, F7, F8, Fz	Fp1, Fp2, F4, F7, Fz	Fp1, Fp2, F7, F8	Fp1, Fp2, F4, F7, F8
T3, T4, T5,T6	T3, T4, T5,T6	T3, T4, T5, T6	T3, T4, T6
O1, O2	O1, O2	O1, O2	O2
Pz	Pz	Cz, Pz	Pz

(b)

B. PSE

An example of the power spectrum measured at Fp1 is shown in Fig 5 where the distribution of signal power is segregated as follows bands: delta (0-4 Hz), theta (4-8 Hz), alpha (8-13 Hz), beta (13-31 Hz), gamma (31-51) and high gamma (51-120 Hz). In this example, it is observed that there is significant power at frequency less than 31 Hz indicating brain activities in the delta, theta, gamma and beta bands.

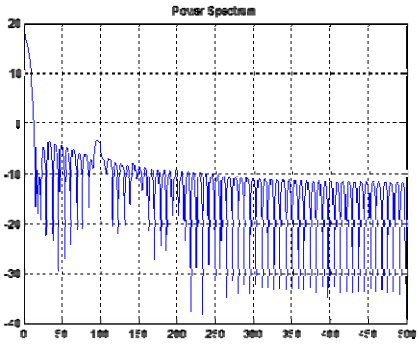

Fig. 5. Example of a power spectrum at Fp1 for a subject while performing a task.

The power spectrum for all the subjects, bands and electrodes are compared between the male and female subjects for all the tasks. Example for Fp1 and Fp2 is shown in Fig. 6 for the delta band. The horizontal axis represents the tasks defined in Fig. 2 while the blue and red bars represent the power measured for the male and female subjects respectively. The Student t-test is used to determine if there is significant difference between the male and female subjects for all task, electrodes and bands.

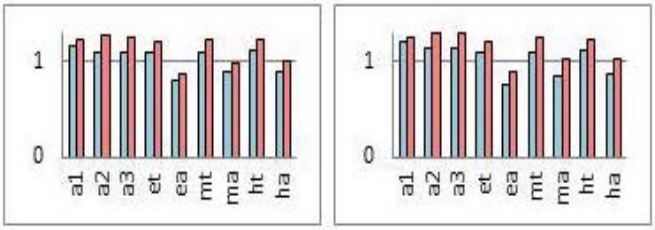

Fig. 6. Comparison between the power in the delta band for Fp1 and Fp2 at state of rest with the various tasks.

The gender differences results based on PSE show changes in control task to problem solving tasks ratio for each of the EEG waveform mentioned. For both male and female subjects, the labelled significant difference can be observed mostly at the frontal regions.

During arithmetic task, the power of delta and theta waves (especially in frontal region) for both gender increased, while the alpha, beta, gamma and high gamma power were decreased. The major waveband used by female subjects was the theta wave while male subjects used theta and beta wavebands majorly during mental arithmetic task.

In contrast, during TOH task, the high gamma wave power for female was increased, while the male subjects showed increased in both the gamma and high gamma power. Moreover, female subjects used their alpha and theta waves majorly, while male subjects used alpha, theta, gamma and high gamma waves majorly.

The different in the change of the wave power as well as the major wavebands involved during the problem solving task were directly related to the subjects' performance. For instance, male subjects who showed best performance in TOH task had correspondingly higher gamma and high gamma power compared to the female subjects

V. Conclusion

On the whole, the term 'differences in problem solving' not only refers to the ability of one to solve a problem, but also refers to an individual way of thinking or thinking pattern or strategy (in a given time) to solve an encountered problem to reach the wanted goal. Especially when dealing with a close-ended solution problem, often people will reach the same solution for the problem regardless of their gender or background. However, the way they solve it or the time taken for someone to solve the same problem may differ from one to another.

Hence, the study of gender differences in problem solving in this study has been done by comparing the performance of subjects based on the gender group (male and female. The EEG data analysis, like PDC and PSE is useful to see the relation between the subjects' performance and the different part of brain regions involves (as shown in PDC results) and wave power distribution (as shown in PSE results).

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